

CN Unit 2 Question Bank — Data Link Layer & MAC

Computer Networks (CN) · Unit 2 · Data Link Layer & MAC · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Write short notes on Framing.
2. Explain Error Detection with a neat diagram and a suitable example.
3. What is meant by Hamming Code? Discuss its advantages and limitations.
4. Explain Sliding Window Protocols with a neat diagram and a suitable example.
5. Write short notes on ALOHA and CSMA/CD.
6. Define Ethernet and Switching. Derive or justify the key properties involved.

Part B — Long answer (10 marks)

1. Discuss the role of Framing in Data Link Layer & MAC.
2. Define Error Detection. Derive or justify the key properties involved.
3. Define Hamming Code. Derive or justify the key properties involved.
4. Compare Sliding Window Protocols with Framing. When would you prefer one over the other?
5. Define ALOHA and CSMA/CD. Derive or justify the key properties involved.
6. Write short notes on Ethernet and Switching.
7. Illustrate Framing with a worked example.
8. Define Error Detection. Derive or justify the key properties involved.
9. Write short notes on Hamming Code.
10. Write short notes on Sliding Window Protocols.
11. Illustrate ALOHA and CSMA/CD with a worked example.
12. Write short notes on Ethernet and Switching.